

Immune aging biomarkers for clinical trials

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Aging profoundly remodels the immune system, impairing defense, repair and homeostatic function across tissues. Because the immune system operates in every organ, its deterioration has been proposed to drive or exacerbate systemic dysfunction and accelerate overall biological aging, making it an attractive biomarker and target for geroscience-guided trials. Despite this central role, there is no consensus on how to quantify immune aging, especially in clinical trials. Here, we establish a translational framework to identify immune aging biomarkers for this purpose. We define five evaluation criteria for immune aging biomarkers and apply these to candidate biomarkers, discussing their utility in the context of a major international healthspan competition, XPRIZE Healthspan. Metrics encapsulating multidimensional aspects of immune function, inflammaging scores and functional assays performed best against our selection criteria. Finally, we identify promising emerging measures, together with critical gaps that must be addressed to develop reliable, predictive biomarkers of human immune competence. Our framework provides a coherent path toward actionable and clinically meaningful immune aging biomarkers capable of quantifying immune fitness and resilience, and accelerating the clinical translation of geroscience-guided interventions.

Aging results from the failure of a complex array of molecular, cellular, phenotypic and functional mechanisms contributing to the exponential increase in the risk of chronic diseases, disability and death¹. As a shared root cause of many pathologies, biological aging represents a unique opportunity for intervention: by targeting its underlying molecular and cellular processes, the onset of multiple chronic diseases could be simultaneously delayed or prevented², with multifactorial downstream effects on healthspan, the period of life spent in good health and free of disability.

Novel drugs, biologics and lifestyle approaches targeting aging processes are actively being developed and advanced to clinical trials. One prominent example is the US\$101 million global incentivized philanthropic competition, XPRIZE Healthspan, which challenges

teams to identify therapeutics that improve function across multiple domains—muscle, cognition and immune—in 1-year phase 2 clinical trials³. Immune aging (see Box 1 for key definitions) represents a tractable and functionally informative domain, with broad relevance for clinical trials as a potential driver of aging^{4,5}. Its clinical and societal relevance was further underscored by the SARS-CoV-2 pandemic, accelerating interest in strategies to preserve immune competence and vaccine responses in older adults^{6,7}. However, unlike physical and cognitive performance, which are routinely assessed using established, standardized functional tests, immune aging lacks accepted metrics that assess its capacity to respond to infection or other insults, thereby preventing both acute and long-term adverse health outcomes⁸. Thus, there is an urgent need to identify and prioritize immune performance biomarkers

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BOX 1

Key definitions relevant to immune aging

Immune aging. The broad, age-associated remodeling of the immune system across cell types, tissues and molecular pathways. Rather than a single dysfunctional state, immune aging encompasses coordinated changes in both innate and adaptive immunity, including impaired generation of new immune cells, altered cell composition, reduced repertoire diversity, functional decline, chronic low-grade inflammation and maladaptive activation. Because the term ‘immunosenescence’ has historically been used inconsistently and is sometimes conflated with cellular senescence or exhaustion, we use ‘immune aging’ as the broader framework describing the progressive loss and dysregulation of immune fitness and immune resilience with age^{133–137}.

Immune fitness. The functional capacity of the immune system to detect, respond to and regulate challenges, encompassing molecular, cellular and systemic attributes determining how effectively the immune system mounts protective responses while avoiding excessive or chronic activation. Thus, immune fitness reflects the ‘preparedness’ or potential performance of immunity at a given time and is shaped by age, genetics, environment and prior immune experiences^{138,139}.

Immune resilience. The immune system’s dynamic capacity to reestablish functional balance when challenged by internal or external stressors. While biological resilience in aging is often defined as recovery to a prior baseline, in the immune system such a return may be maladaptive, as it could entail loss of memory or reduced capacity to combat pathogens. Instead, immune resilience reflects an allostatic process, whereby the system establishes a new personal baseline that preserves protection, immunological memory and surveillance while avoiding inflammation, autoimmunity or immune exhaustion^{140,141}.

Immune cell senescence. The irreversible cell cycle arrest of immune cells during which they upregulate senescence-related genes and may acquire a SASP, involving the secretion of pro-inflammatory cytokines (for example, IL-6 and IL-8), chemokines and proteases, which can contribute to chronic inflammation and tissue dysfunction¹⁴²; these may vary across cell types, as recently discussed elsewhere^{143,144}.

T cell exhaustion. A distinct differentiation state of T cell dysfunction arising during chronic antigen stimulation, such as in persistent infections or cancer. Exhausted T cells display progressive loss of effector functions, cytokine dysregulation, persistent expression of inhibitory receptors and undergo transcriptional and epigenetic reprogramming that stabilizes their dysfunctional state^{145,146}. T cell exhaustion is not equivalent to immune aging or cellular senescence. In aging, T cells may instead exhibit a mixture of reduced function, clonal expansion, terminal differentiation, altered signaling and, in some cases, exhaustion-like features, but these processes should not be treated as interchangeable.

Inflammaging. A chronic, low-grade, systemic inflammatory state that accompanies advancing age, particularly in industrialized settings, and may differ by sex^{59,87,147,148}. It is characterized by altered basal levels of circulating inflammatory mediators and is associated with frailty, multimorbidity and multiple age-related diseases^{15,17}. At the same time, age-associated inflammatory responses may not be uniformly detrimental: they may reflect context-dependent compensatory or adaptive responses to accumulated molecular damage, tissue stress or impaired homeostatic mechanisms. Thus, inflammaging should be viewed as a complex and context-dependent feature of aging, with both pathological and potentially adaptive dimensions¹⁴⁹.

for clinical trials attempting to boost immune function and counteract the effects of immune aging, and to establish reference data of trajectories of longitudinal variation in immune function over the life course.

Here, we present a conceptual framework for the selection of immune aging biomarkers suitable for randomized controlled clinical trials in geroscience, developed through structured literature review, iterative refinement and guided expert workshops with a multidisciplinary Immune Aging Biomarkers workgroup (comprising immunologists, clinical trialists, geroscientists and industry stakeholders). The framework was developed in the context of the XPRIZE Healthspan competition, which provided a concrete and real-world trial setting against which to test and refine biomarker selection criteria. We delineate dimensions of immune aging relevant to biomarkers, establish key selection criteria for immune aging biomarkers in clinical trials, and critically evaluate existing candidates against these criteria. Finally, we map strategic directions for biomarker development, leveraging existing or new data collections, emerging technologies and AI-enabled analytics. The result is an evidence-based prioritization of candidate immune aging biomarkers for early-stage trials targeting the underlying biological mechanisms of aging—termed here geroscience-guided trials—alongside a translational roadmap for biomarker development to support the next generation of clinical studies on aging.

Immune aging: dimensions, classification and systemic interactions

The immune system interfaces with every organ and tissue, sustaining immune surveillance and defense, tissue homeostasis and repair.

Aging profoundly remodels the immune system, altering and in many cases undermining many of these functions^{9–11}. The observed progressive decline in pathogen defense, vaccine responsiveness and regulatory balance, alongside persistent low-grade inflammation, is collectively termed immune aging. These changes operate across interconnected molecular, cellular and organ levels¹⁰, and their systemic consequences—higher infection susceptibility, impaired cancer surveillance and the emergence of autoimmunity (driven by thymic involution, impaired tolerance mechanisms and chronic inflammatory remodeling)—contribute to chronic disease, frailty, cognitive decline and mortality^{12–17} (Fig. 1). Although not intended to provide a full overview of all aspects of immune aging (recently reviewed in refs. 10,18,19), we provide a brief overview of relevant dimensions for clinical immune aging biomarkers.

At the organ level, the thymus undergoes progressive atrophy beginning in early life (a process of ‘thymic involution’), during which functional thymic tissue is gradually replaced by fat and fibrotic tissue. This reduces the production of naive T cells, diminishes T cell diversity and heightens susceptibility to infections^{20–23}. Bone marrow hematopoietic stem cells exhibit aberrant proliferation, self-renewal and myeloid differentiation bias with age, often interlinked with clonal expansion of single hematopoietic stem cell clones—known as clonal hematopoiesis of indeterminate potential, or CHIP^{24,25}. CHIP is common in aging and is associated with increased risk of cardiovascular disease and hematologic cancers²⁶, although with reduced risk of some conditions such as Alzheimer’s disease²⁷. Age-related atrophy and structural remodeling of lymph nodes further compromise adaptive immunity

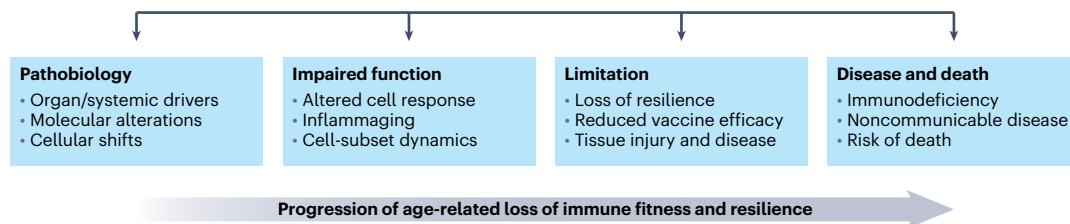


Fig. 1 | Immune aging dimensions. The intricate interplay of immune aging dimensions is essential to consider when developing effective and clinically relevant immune aging biomarkers. These dimensions include: (1) pathobiology, encompassing molecular or high-dimensional signatures that reveal early organs or cellular damage; (2) functional impairments, represented by composite

biomarker profiles used to identify altered cell responses in the absence of clinical manifestations; (3) clinical limitations, evident as increased susceptibility to infection, tissue damage or reduced vaccine efficacy; and (4) disease or mortality, including immunosuppression, immune-related noncommunicable diseases or infection-associated death.

by disrupting the microenvironment necessary for effective immune cell communication and activation²⁸.

These organ-level changes drive and reinforce cellular shifts across the immune landscape. While myeloid cell production increases with age, myeloid cell function is frequently compromised^{29,30}. Single-cell analyses and T cell antigen receptor repertoire profiling reveal a reduction in naive T cells and accumulation of dysfunctional memory T cells, terminally differentiated and clonally expanded T cells, contributing to a less diverse and less responsive adaptive immune system^{31–34}. Concurrently, self-renewing resident tissue macrophage populations decline with age across multiple organs^{35–37}. In mice, loss and functional impairment of these cells have been implicated in defective tissue repair, impaired protection against infections and age-related diseases including fibrosis and cancer³⁵, with emerging human data suggesting similar age-related depletion³⁸. This cellular remodeling provides a mechanistic bridge between chronic inflammation and organ-specific vulnerability.

At the molecular level, epigenetic alterations—including changes in DNA methylation and histone modifications—represent a key interface between environmental exposures and immune gene regulation^{39,40}, and have been proposed to underlie the proliferative exhaustion and dysfunction of immune stem cells^{41,42}. Sex-specific chromosomal alterations also contribute: loss of the Y chromosome in hematopoietic cells of aging men impairs immune surveillance and promotes inflammation, while X-chromosome skewing in women may alter gene dosage and contribute to hematopoietic dysfunction^{43,44}. Telomere attrition is a key trigger of cellular senescence in leukocytes^{33,45}, although recent evidence suggests telomere length may be partly genetically determined, chromosome specific⁴⁶ and stable over the life course in some species⁴⁷, warranting further investigation. Underlying many of these changes is persistent, age-related pro-inflammatory signaling (‘inflammaging’; Box 1)—alongside mitochondrial dysfunction and defective mitochondrial turnover^{48,49}. Importantly, aged immune cells propagate inflammatory and senescent signals to surrounding tissues, amplifying dysfunction both locally and systemically.

Declines in immune function thus arise from interconnected changes, with each dimension influencing others: for instance, thymic involution directly causes a decrease in newly created naive T cells, which in turn can alter the landscape of circulating inflammatory markers⁵⁰. This intricate interplay is essential to consider when developing effective and clinically relevant immune aging biomarkers, as the most robust and meaningful indicators will be those that capture the connections ranging from molecular alterations to clinical outcomes^{51,52} (Fig. 1). These domains are interdependent and bidirectional. They also represent hierarchical levels of measurement for aging-targeted clinical studies, with salience, cost and observation time depending on which dimension is chosen as the outcome measure⁵³. Importantly, given the dynamic nature of the immune system, informative assessments should capture longitudinal dynamics of immune fitness and resilience (Box 1), including response kinetics, memory formation

and recovery following controlled challenges such as vaccination. Ultimately, large or long-term clinical trials could evaluate interventions by their capacity to enhance immune fitness and resilience, measured by reduced infection incidence, improved vaccine efficacy, faster recovery or lower infection-related mortality (for example, influenza or SARS-CoV-2) in older adults.

Criteria for immune aging biomarkers

Existing biomarker evaluation frameworks provide essential foundational criteria applicable to immune aging^{54–56}, including feasibility for repeat measurement without incurring undue risk, compatibility with trial logistics (for example, cost, scalability, processing capacity), high analytic reliability and robustness to stochastic fluctuations unless biologically justified. More broadly, biomarkers of aging are defined as quantitative parameters that, either alone or in a composite, predict biological age—reflected in the accumulation of molecular and cellular damage—and ideally its changes in response to interventions⁵⁴. The **Biomarkers of Aging Consortium** established consensus on key working terminology and evaluation criteria to support the translational use of such biomarkers^{54–56}. Additional context-specific criteria can assess whether a biomarker reflects underlying biology (plausibility), responds to interventions targeting biological aging and, critically, whether changes in the biomarker reflect changes in biological aging and age-related disease risk, qualifying it as a surrogate endpoint⁵⁴.

However, immune aging presents additional challenges that general frameworks do not fully address. Quantifying immune aging is inherently complex, requiring the integration of diverse, often interdependent parameters spanning innate and adaptive immunity, inflammation and tissue-level dysfunction, as discussed above (see also Fig. 1). Moreover, immune aging biomarkers should ultimately capture not just static immune features but also the system’s capacity to recognize, respond to and eliminate pathogens or damaged cells while maintaining tolerance and tissue homeostasis—and critically, its ability to do so adaptively under challenge. Importantly, inclusion of a candidate biomarker in this framework does not imply that its direct modulation is sufficient to reverse immune aging, since biomarker utility for measurement and trial readout is distinct from mechanistic causality or therapeutic actionability.

We therefore propose five criteria tailored specifically to biomarkers of immune aging in clinical trials: immune relevance, temporal association, internal validity, predictive value and responsiveness (Box 2). These criteria address distinct biological, technical and translational challenges of measuring immune aging, and build on foundational considerations. Key among these is feasibility in each context of use, which will necessarily vary across settings depending on infrastructure and expertise. For example, some sites routinely process cryopreserved peripheral blood mononuclear cells, perform flow cytometry or store fixed cells from whole blood for reduced flow panels, while resources may prohibit such analyses at other sites. Such differences should not disqualify otherwise strong trial designs, and feasibility must

BOX 2

Criteria for immune aging biomarkers in clinical trials

Immune relevance: Mechanistically linked to immune function, ideally capturing multiple aspects (for example, innate and adaptive components) rather than a single pathway, and reflecting general processes of immune function or aging rather than being restricted to a specific disease, toxin or environmental exposure.

Temporal association: Sensitive to biologically meaningful, within-person changes and trajectories over time, consistent with immune aging processes.

Internal validity: Measurable and reproducible according to established performance metrics (for example, test–retest precision, analytical sensitivity (limit of detection), specificity, linearity, dynamic range, stability during handling and transport). Minimally influenced by transient physiological fluctuations unless biologically justified.

Predictive value: Predictive of risk of mortality and multiple age-related outcomes. If a change in the biomarker predicts a change in the outcome, this can potentially become a surrogate endpoint.

Responsiveness: Demonstrates biologically consistent change in response to positive or negative exposures, such as therapeutic intervention, stressors or loss of health, and capable of changing within the expected time frame of an intervention or study.

therefore be interpreted flexibly and transparently. The target population is another foundational consideration, since generalizability and interpretability may differ across age groups, disease states, genetic backgrounds or trial designs.

For geroscience-guided trials with immune function as a target or trial endpoint, changes in immune aging biomarkers should correspond both to improved immune function and prediction of clinically meaningful outcomes, such as delayed onset or severity of acute illness or mitigation of the accumulation, clinical evolution and functional consequences of chronic diseases or disability. By design, immune aging biomarkers may hold particular promise as a mechanistically interpretable and clinically actionable subgroup of biomarkers of aging. They may have a key advantage over many existing biomarkers⁵⁶ as declines in immune function have been linked to both acute and chronic morbidity and mortality^{57,58}, supporting potential causal relevance for aging processes.

A key consideration is that the immune system operates allostatically rather than purely homeostatically: it sustains stability by continuously adapting to challenges and retaining immunological memory, not by reverting to a single baseline state. This has direct implications for biomarker design. Perturbation models, such as vaccine response, can reveal resilience that is not apparent at rest, analogous to orthostatic challenge in blood pressure or postprandial glucose in metabolism. Immune aging biomarkers must therefore capture not only static features but also the capacity for dynamic response across immune functions.

At the same time, inflammation-based biomarkers require particularly careful interpretation in longevity clinical trials, as they are highly responsive to factors unrelated to age, including comorbidities, acute infections, obesity, medications and other determinants of immune state. Recent evidence also suggests that inflammaging signatures are not universal across populations, and that inflammatory profiles associated with aging may differ from those driven by infection or

other environmental and clinical exposures⁵⁹. For example, data from healthy adults suggest that key age-associated immune trajectories can be distinguished from both latent cytomegalovirus (CMV)-associated remodeling and overt systemic inflammation, underscoring the need to interpret inflammation-based biomarkers in the context of relevant clinical and infectious covariates⁶⁰. These factors should therefore be considered explicitly in longevity trial design and treated as key covariates in analysis; in addition, longitudinal within-person sampling may be especially valuable for distinguishing age-related immune change from transient or context-dependent inflammatory variation.

Finally, although a detailed operational framework is beyond the scope of this Perspective, we emphasize that integration of immune aging biomarkers in clinical trial designs requires prespecified minimum standards for biospecimen acquisition and assay performance. While some parameters will necessarily depend on the intervention, biomarker class and analytical platform, studies should at minimum define the primary sample type, standardized processing time windows, storage and transport conditions, acceptable intra-assay and inter-assay variability, and core quality-control measures including sample integrity, internal controls, instrument calibration and batch consistency. The prospective definition and consistent application of these criteria will be essential for ensuring reproducibility, inter-site comparability and longitudinal interpretability of immune aging biomarkers.

Critical evaluation of candidate immune aging biomarkers

Over the past five decades, numerous candidate biomarkers of immune aging have been proposed, yet few have been rigorously evaluated against consistent criteria or validated for clinical use. We compiled a list of existing candidate biomarkers (Table 1; see footnote for selection criteria) including systemic circulatory biomarkers, cell proportion-based biomarkers, DNA methylation or gene expression-based biomarkers, among others. We then qualitatively assessed each candidate against our five criteria (Box 2) using relative scores (high, moderate or low support), informed by available longitudinal or cross-sectional data, reproducibility and evidence of clinical relevance. Below, we provide an overview of the included biomarkers and their assessment.

Overview of current candidate biomarkers

Systemic circulatory biomarkers include soluble factors in blood or plasma such as senescence-associated secretory phenotype (SASP) proteins, inflammatory indices and multiplex cytokine profiles. These biomarkers capture systemic inflammation, often reflecting inflammaging. Given the many methodologies and ease of measurement, these are among the most accessible biomarkers for clinical translation. In this category, we prioritized panel-based and multi-marker approaches, deliberately excluding single-molecule biomarkers that provide limited information on the immune status as a whole and are context dependent (biased by disease states or acute physiological changes). Although systemic circulatory biomarker panels show strong associations with age, they may originate from diverse cell types, making it conceptually difficult to isolate whether they selectively encapsulate immune aging or alterations in other cell types (or an increase in senescent cells). Nevertheless, we consider these biomarkers essential to the immune aging framework since they exert widespread effects on immune regulation and reflect systemic processes that shape immune competence.

Cell proportion-based biomarkers, derived from immunophenotyping (usually flow cytometry), quantify shifts in the abundance of immune subsets, for instance the decline of naive T cells, skewing toward memory and effector populations, and the accumulation of senescent or exhausted subsets. These features primarily reflect adaptive immune aging.

DNA methylation and gene expression-based biomarkers encompass epigenetic clocks or alternative composite algorithms, transcriptional signatures and molecular readouts from sequencing or

Table 1 | Summary and evaluation of established and emerging biomarkers of immune aging

Details ^a		Criteria ^b		Temporal association	Internal validity	Predictiveness	Responsiveness
Biomarker name and details	Category	Dimension	Immune relevance				
SASP Panel ⁷⁶ (GDF15, FAS, OPN, TNFR1, activin A, CCL3, IL-15)	Systemic circulatory biomarkers	Inflammaging	Low	Moderate ($r=0.33-0.75$ with chronological age)	NE	Moderate to strong (aortic stenosis ⁷⁶ , ovarian cancer ⁷⁶ , mobility limitation ⁷⁷ , heart failure ⁷⁷ , all-cause mortality ⁷⁸ , SPPB ³² , 400m walk ⁵² , mild cognitive impairment ⁴⁰ , treatment outcomes in late-life depression ⁸⁹)	Moderate to strong (physical activity ⁷⁹)
V-PLEX pro-inflammatory panel (IL-6, CRP, IFN γ , IL-1 β , IL-2, IL-4, IL-6, IL-8, IL-10, IL-12p70, IL-13 and TNF)	Systemic circulatory biomarkers	Inflammaging	Moderate	Moderate	NE	NE	NE
iAge ⁶⁵ (CXCL9 plus 49 other cytokines)		Inflammaging	Strong ⁶⁵	Strong ⁶⁵	NE	Moderate (SPPB, 5STS in INSPIRE ⁷³) to strong (multimorbidity, immune aging, frailty and cardiovascular aging, infection-related mortality, all-cause mortality, cardiovascular and metabolic disease risk factors ⁷⁴)	Moderate (191-person RCT of oats on CVD and iAge ⁹⁰ ; 42-person trial of plasma exchange ⁹¹)
Glycan clocks (IgG glycosylation)	Systemic circulatory biomarkers	Molecular alterations	Moderate ⁹²	Moderate to strong ⁹³	Strong ⁹⁴	Moderate (mostly risk of disease in case-control studies, not longitudinal cohorts; ischemic stroke ⁹⁵ , mortality in sepsis ⁹⁶ , CVD risk ⁹⁷ , tuberculosis progression ⁹⁸)	Low (selective change with testosterone but not metformin in young male adults ⁹³)
Neutrophil-to-lymphocyte ratios	Cell proportion-based biomarkers	Altered cell subtypes	Strong ⁹⁹	Moderate to strong ⁹⁹ (ratio increases with age in many populations, including general US population, for example, NHANES; strength due to routine measure with complete blood counts in clinical cohorts)	Moderate	Moderate (all-cause mortality ⁹⁹ , cause-specific mortality including respiratory issues (influenza, pneumonia), kidney, and heart disease ⁹⁹ , grip strength ¹⁰⁰ , physical function ¹⁰¹ , short-term mortality after myocardial infarction ¹⁰² , CVD ¹⁰³ , breast cancer ¹⁰⁴ , Alzheimer's disease risk biomarkers ¹⁰⁵)	Moderate (IL-6 inhibitor for atherosclerosis lowers neutrophil-to-lymphocyte ratio)
Immune risk phenotype (IRS; CD4/CD8 ratio)	Cell proportion-based biomarkers	Altered cell subtypes	Moderate (ratio <1 reflects immune aging)	Moderate (most studies in persons aging with HIV, not general aging population)	NE	Moderate (positive associations with CVD risk ¹⁰⁶ and cancer survival ¹⁰⁷ , negative associations with mortality in CMV-negative women ¹⁰⁸ , association with mortality ¹⁰⁹)	Moderate (cancer or HIV only)
Aging related immune phenotype (AIRP) ¹⁰⁻¹¹² (CD4 and CD8: naive T cells/central memory T cells + effector memory T cells + effector T cells))	Cell proportion-based biomarkers	Altered cell subtypes	Strong ¹¹²	Strong ^{111,112}	NE	Strong (all-cause mortality, multimorbidity in Health & Retirement Study ¹¹² ; overall survival in Framingham Heart Study ¹¹⁰)	NE
InflammAge ^{75,113} (DNA methylation proxy of chronic inflammation)	DNA methylation and gene expression-based biomarkers	Inflammaging	Low to moderate (built using DNA methylation proxies from 125 immune-related proteins, such as CRP, IL-6 or CXCL10)	Moderate (limited cohorts)	Moderate to strong (validated as a laboratory-developed test according to CLIA standards)	Moderate (CVD-related ¹¹⁵ , electronic health record associations in Generation Scotland cohort for all-cause mortality, disease outcomes, immune aging ¹¹⁶)	Low (reduced in 80 healthy individuals after supplement only in group with InflammAge acceleration at baseline ¹¹⁴ ; no significant change in 50 Alzheimer's patients with anti-inflammatory agent ¹¹⁵)

Table 1 (continued) | Summary and evaluation of established and emerging biomarkers of immune aging

Details ^a		Criteria ^b					
Biomarker name and details	Category	Dimension	Immune relevance	Temporal association	Internal validity	Predictiveness	Responsiveness
TREC quantification (to estimate recent thymic emigrants ¹⁶)	DNA methylation and gene expression-based biomarkers	Organ and systemic drivers	Moderate (good in theory, some quantitative issues in lab ¹⁶)	Moderate (general population ¹⁷ , after allogeneic HSCT ¹⁸ , cardiac surgery >50years—weak associations with age ¹⁹)	Moderate (TREC-based assays are inherently insensitive to subtle changes in thymic output ¹¹⁹)	Low to moderate (best in specific patient populations, like neonates, AML, HSCT recipients)	NE for gerotherapeutics (only immune deficiency disorders and cancer)
IMM-AGE ⁶⁵ (gene expression, cell proportions)	Multimodal biomarkers	Altered cell subtypes	Strong ⁶⁵	Strong ⁶⁵	NE	Strong (CVD ⁶⁵ , all-cause mortality ⁶⁵ , sepsis ⁷⁰ , cardiorespiratory fitness ⁷¹ , work ability ⁷²)	NE
Unified health metric ²⁰ (cell counts, transcriptomics, proteins)	Multimodal biomarkers	Molecular alterations	Strong (calibrated in persons with 22 monogenic conditions affecting key immunological pathways or healthy controls ²⁰)	Moderate (r=−0.56 in a single cohort)	NE	Moderate (predicted age-dependent antibody responses to immunizations with different vaccines ²⁰)	NE
Infection rates (response to pathogen exposure)	Functional immune biomarkers	Immune fitness	Strong	Strong (influenza by US Centers for Disease Control and Prevention ⁶¹ , COVID-19 ⁶ , West Nile virus ⁸² and other infections ⁶³)	NE existing clinical outcome assessment (best)	NE clinical outcome assessment: also predicting multimorbidity and higher healthcare costs ²¹	NE (few examples, with mixed results — mTOR inhibitor phase 2 trial was positive ⁸⁵ but phase 3 trial failed to meet FDA outcome ⁸⁴)
Antibody response to vaccine (cellular immunity)	Functional immune biomarkers	Vaccine efficacy	Strong ¹²²	Strong (for example, SARS-CoV-2 vaccine ¹²³ , influenza antibody titers and aging ¹²⁴)	Strong	Moderate (association of seroconversion/seroreversion with frailty status ²⁵ , COVID-19 severity and failed response in persons with multimorbidity ²⁶)	Moderate (+/−; phase 2 showed a 20% increase ⁸⁵ ; metformin pilot negative ¹²⁷)
Rapid high-throughput T cells and monocyte response to activation signals (ex vivo activation of cellular immunity)	Promising biomarker candidates yet to be developed	Altered cell response	Strong ^{125,129}	Moderate to strong ¹³⁰	NE	NE	NE
Loss of resident tissue macrophages	Promising biomarker candidates yet to be developed	Altered cell response	Strong	Moderate to strong ^{315,131,132}	NE	NE	NE
Immune cell mitochondrial dysfunction (respiratory capacity)	Promising biomarker candidates yet to be developed	Molecular alterations	Strong	NE	NE	NE	NE
Lymph node or bone marrow structure on imaging	Promising biomarker candidates yet to be developed	Organ and systemic drivers	Strong	Moderate (thymic volume ²⁷)	NE	NE	Low (thymic volume ²⁷)

^aCandidate biomarkers were included if they met the following three criteria: (1) specificity and relevance to at least one dimension of immune aging, (2) sufficient documentation to enable standardized measurement, (3) and measurability from accessible sample (blood or saliva) or data types using standard laboratory techniques (for example, flow cytometry, ELISAs, sequencing-based or array-based assays, MRI). ^bEvidence strength is graded as low, low to moderate, moderate, moderate to strong, or strong. CVD, cardiovascular disease; HSCT, hematopoietic stem cell transplantation; NA, not applicable; NE, not evaluated or reported; RCT, randomized controlled trial; TREC, T cell antigen receptor excision circle.

high-dimensional profiling. These measures capture epigenetic alterations and transcriptional remodeling associated with aging, but the evidence for specific associations with immune fitness and resilience remains limited. Multimodal biomarkers combine diverse data layers (for example, proteomic, transcriptomic, cellular composition) into unified models that may better capture the multidimensional nature of immune aging.

Frontier technology biomarkers encompass emerging technologies and metrics that have demonstrated promise in research but have not yet been developed into standardized assays. For instance, such biomarkers may include metrics describing tissue-level immune aging, including lymph node, thymic and bone marrow architecture, cell function in response to pathogens, cell-intrinsic capacity (for example, mitochondrial metabolism) and responses to ex vivo stimulation. Although these are promising approaches to directly assess central drivers of immune resilience and decline, they currently lack concrete assays standardized across labs and remain difficult to scale in clinical or large-cohort settings. It is noteworthy that molecular or composite biomarkers have been developed to serve as proxies of functional performance; however, this remains a currently underdeveloped area requiring further refinement and standardization for integration into biomarker panels that can be broadly implemented in clinical and translational settings.

We note that beyond these biomarkers, many other molecular or cellular features have been associated with immune aging, ranging from changes in gene expression levels to chromosomal loss^{61–64}. While biologically informative, these associations have rarely been tested in prospective studies and therefore cannot yet be considered immune aging biomarkers.

Assessment of candidate biomarkers

The candidate biomarkers discussed above vary substantially in biological specificity and technical maturity, as well as the availability of data to evaluate specific aspects of translational and clinical readiness. Below, we summarize key insights from our evaluation (Table 1) of these biomarkers against each of our five proposed criteria for immune aging biomarkers, highlighting where evidence is strong, where it is emerging and where most critical gaps remain.

Immune relevance. Current biomarkers vary widely in their relevance to immune aging. Several frequently reported biomarkers are associated with age but lack specificity to immune aging, being strongly influenced by infections, cancer, cardiovascular disease and autoimmunity. Others primarily capture late-stage decline rather than early functional impairment: for instance, the loss of naive CD8⁺ T cells, the decline in newly created CD4⁺ T cells or the accumulation of senescent T cells usually becomes evident only after substantial immune deterioration. At the opposite end of the spectrum lie functional assays such as ex vivo stimulation or vaccination challenges, which provide highly specific readouts of immune resilience and function. In between these extremes, omics approaches (single-cell omics, high-dimensional cytometry, proteomics, transcriptomics and DNA methylation profiling) represent an emerging class of biomarkers that can be used to train algorithms to capture diverse immune aging processes; examples include the high-dimensional trajectory of immune aging (IMM-AGE)⁶⁵ and the inflammatory clock of aging (iAge)⁶⁶.

Temporal association. Few biomarkers have been rigorously evaluated in longitudinal settings, making it difficult to distinguish progressive immune aging from short-term variation and interindividual differences observed in cross-sectional studies. Longitudinal immune challenges such as vaccination arguably provide the most direct reflection of temporal variation and resilience. Without temporal resolution, it is unclear whether a given parameter reflects progressive aging, compensatory responses or short-lived perturbations.

Internal validity. Biomarkers in Table 1 differ widely in reproducibility and standardization. Measures such as cytokine assays can be rapidly reproduced, and subcomponents may already have undergone standardization and accreditation for clinical testing purposes. By contrast, many of the most conceptually powerful and relevant biomarkers for immune aging (for example, ex vivo stimulation assays) do not have standardized assays and protocols at all: they remain bespoke research tools without established performance metrics, making cohort-wide validation or cross-cohort comparison difficult. Ideally, assays would report a coefficient of variation of less than 10% for technical replicates, and test–retest correlations of at least $r > 0.70$ over a period of weeks. Yet, across most biomarker classes, test–retest reliability, sensitivity and stability under real-world handling conditions remain largely underreported. Further complicating standard validity required for bioassay development, reliability assessments assume the assay measures a stable trait, but the immune system, by its nature, fluctuates in response to external exposures and perturbations.

Predictive value. Although several hallmarks of immune aging have been described^{33,67,68}, few candidate biomarkers have been prospectively validated using standardized assays as predictors of morbidity, multimorbidity or mortality. This complicates their evaluation and limits their adoption in clinical trials. IMM-AGE⁶⁵ and iAge⁶⁶ are among the most advanced: both have been validated in independent cohorts. IMM-AGE (cytometry scores or -omic proxies thereof) captures a conserved immune developmental process identified through longitudinal data, quantifiable by multiple omics-based immunoprofiling technologies, while iAge is a summary measure of inflammaging based on patterns of circulating age-related inflammatory cytokines and chemokines. IMM-AGE has been demonstrated to be elevated during the earliest stages of disease in rheumatoid arthritis⁶⁹ and in trauma patients at risk of sepsis⁷⁰, and high IMM-AGE score predicted low cardiorespiratory fitness⁷¹ and reduced ability to work in independent populations⁷². Higher iAge values are associated with worse physical fitness⁷³ and exhibit associations with several age-related outcomes, including frailty, multimorbidity and all-cause mortality, among others⁷⁴. Together with other candidates, including cytokine panels, epigenetic biomarkers of inflammation (for example ref. 75) and certain multi-omic signatures, they show promise but require broader replication to rapidly transition from research tools to robust and informative clinical trial endpoints.

Responsiveness. Responsiveness to interventions (sensitivity to change) is the least developed dimension of immune aging biomarkers. Some markers (for example, cytokines) are known to shift with acute exposures, but these changes may reflect adaptive or compensatory processes rather than immune aging. Without intervention studies, it remains difficult to know which biomarkers consistently respond in biologically meaningful ways, over which time frames, with what degree of stability and whether they can serve as surrogate endpoints.

Importantly, most immune aging biomarkers discussed here should be considered pharmacodynamic or risk biomarkers rather than validated surrogate endpoints. Although several measures are associated with morbidity and mortality, qualification as a surrogate endpoint would require evidence across multiple independent interventions demonstrating that changes in the biomarker consistently mediate clinical benefit. This distinction is critical to avoid overinterpretation of short-term biomarker changes and highlights the need for longitudinal validation, linking modulation of immune aging biomarkers to clinically meaningful health outcomes. Emerging efforts are explicitly designed to address these evidentiary gaps. For example, the [Proactive Solutions for Prolonging Resilience](#) program within ARPA-H will fund validation efforts for biomarkers of aging, and the Biomarkers of Aging Consortium and the [Human Immunome Project](#) are pushing toward validation of immune aging biomarkers as well.

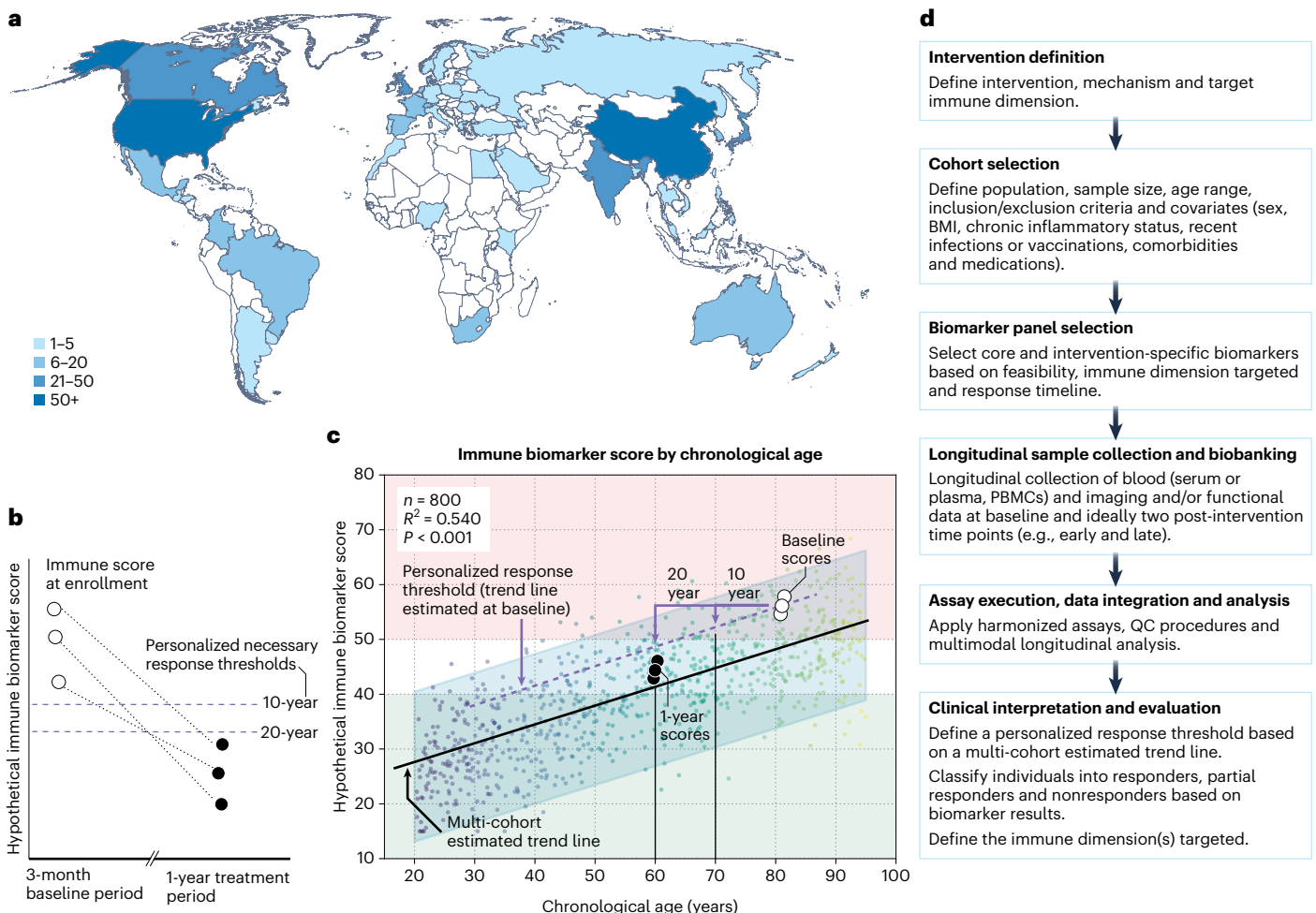


Fig. 2 | XPRIZE Healthspan context of use. **a**, Broad global representation. Heat map of the global distribution of registered XPRIZE Healthspan teams in 2025 by country. In this competition, competing teams will test diverse middle-aged and older adult populations across sites with varying infrastructure, resources, expertise and capabilities in global multicenter trials. The broad geographical representation emphasizes the need for easily standardized protocols for biospecimen collection and processing: complex processing may increase error and variability and therefore assays with simple collection SOPs are prioritized. Moreover, access to specialized equipment or trained personnel for handling live cells, viruses, bacteria or pathogens cannot be assumed. Ideally, assays should be compatible with frozen specimens shipped to central laboratories. **b**, Within-person benchmarking: example of immune biomarker scoring. Endpoints within XPRIZE trials are assessed in aggregate single-cross-over trials with repeat baseline assessments used as the untreated within-person condition. Specifically, three untreated baseline assessments and three follow-up assessments at 9, 11 and 12 months will be used to evaluate change following active intervention versus placebo/standard-of-care, requiring the use of assays

that: (1) are feasible for serial monthly assessments; (2) have high technical validity, standardized procedures and low intraindividual variability; and (3) are continuous measures reflecting functional improvement, not event counts or risk attenuation. **c**, Reference thresholds for ‘meaningful’ change: hypothetical development of personalized reference thresholding for 10-year and 20-year change. Judges will use personalized age-stratified and sex-stratified response thresholds to evaluate within-person change from baseline to 10–12 months. Participants exceeding their threshold (per age/sex) over three baseline measures are deemed ‘responders’. This requires generalizable reference cohort data with age-appropriate and sex-appropriate norms, ideally from longitudinal data informing 10-, 15- and 20-year changes. **d**, Decision workflow for inclusion of immune age biomarkers within clinical trials such as XPRIZE Healthspan. Selection criteria are context-dependent, and trialists should consider factors such as intervention, cohort or population, panel selection and feasibility, longitudinal collection time points, assay execution and data analysis, and interpretation of findings. PBMs, peripheral blood mononuclear cells; QC, quality control.

Immune aging in clinical trials: the XPRIZE Healthspan example

The XPRIZE Healthspan competition provides a timely and concrete opportunity to translate the above framework into practice. The competition challenges teams to demonstrate, within a one-year phase 2 clinical trial, that a single or combination of therapeutics can restore function lost to age-related degradation across three organ systems and domains—muscle, cognitive and immune—emerging from deliberate scientific engagement with an endpoints committee and advisory board. Importantly, the therapeutic treatment’s beneficial effects should yield a plurality of effects across these systems and must demonstrate improvements rather than attenuation of decline. Teams test

their interventions at their own clinical sites using common protocols and standard operating procedures (SOPs), while relying on a centralized data coordinating center and laboratories for data management and assay harmonization. Several unique features of XPRIZE Healthspan’s design, including decentralized global execution, heterogeneous populations and infrastructure, and a one-year time horizon—create specific constraints that directly shaped biomarker selection (Fig. 2a–c); however, the underlying framework and evaluation criteria are intended to be broadly applicable to geroscience-guided clinical trials. In this section we describe how we applied our five selection criteria (Box 2) to identify candidate immune aging biomarkers for this competition.

Selected biomarkers for the XPRIZE Healthspan competition

No single metric of immune aging is sufficient on its own, and the selection was guided by the need for nonredundant measures that together capture complementary dimensions of immune aging, while remaining feasible in a global decentralized trial. Based on expert consensus and the application of the five criteria outlined above, two nonredundant, prespecified immune aging biomarkers were identified: iAge⁶⁶ (or similar multidimensional inflammaging scores), representing an inflammatory aging clock, and IMM-AGE, an immune aging metric based on high-resolution immune cell composition (or proxies thereof)⁶⁵, both described in sections above. Together, these biomarkers are associated with chronological age across populations, linked to mortality (all-cause or immune-related), noncommunicable chronic disease, geriatric syndromes and multimorbidity, and have preliminary evidence of responsiveness to interventions. Critically for trial feasibility, both can be measured serially with acceptable internal validity; biospecimen collection and processing can be standardized across sites, and both show potential for automation that could accelerate throughput.

Multiplexed panels were biologically promising and met many of the selection criteria, more so than individual inflammatory proteins (for example, interleukin-6 (IL-6), tumor necrosis factor (TNF) or C-reactive protein (CRP)) that we consider to inadequately capture the complex interacting systems that comprise immune fitness and are heavily compounded by illness, disease states and nonimmune processes. For instance, the Mayo Clinic SASP Panel—derived by screening the secretory profile of induced senescence—is highly predictive of many relevant functional and disease outcomes^{76–80} and responsive to exercise in older adults⁷⁹. However, panels of this kind typically require interpretation of change across individual biomarkers (for example, growth differentiating factor-15 (GDF15), TNF receptor 1 (TNFR1) or activin A) rather than a single multidimensional score, introducing practical challenges for adjudicating responders versus nonresponders in the XPRIZE trial context.

The missing link: a dynamic cell response proxy

We propose that the most important biomarker is one that does not yet exist in clinical trial-ready form: a robust, standardized assessment of dynamic immune response to challenge. Static measures of immune composition or inflammatory tone capture baseline state of the immune system but cannot directly assess its capacity to mount, sustain and resolve an immune response—the core functional property that declines with aging and that interventions should seek to restore. Ideally, trials would capture immune resilience through clinical measures such as rates of infection with influenza, SARS-CoV-2 or other relevant pathogens^{6,81–83}, which are immediately meaningful and reflect real-world immune competence. Although such data should be collected where possible, the sample sizes and time horizons required to demonstrate a treatment effect on infection rates exceed the scope of many early-stage trials, and this is challenging for decentralized global trials that experience highly variable exposures or risks across clinical sites. Induced challenge measures, such as antibody response to seasonal vaccine or ex vivo cell response to novel pathogen exposure, offer a more controlled alternative and have demonstrated sensitivity to intervention. Antibody responses to vaccination have shown improvements of up to 20% in trials of geroprotectors like mTOR inhibitors, although generalizability remains an open question as this trial was conducted in a relatively homogeneous New Zealand population^{84,85}. Standardizing assays like vaccine efficacy, antibody response and ex vivo cell stimulation presents additional logistical challenges given the unique constraints and complexity of the global competition.

We therefore challenge the field to develop a proxy biomarker that reflects individual immune fitness and resilience in a format suitable for global trials, ideally measurable in banked biospecimens, meeting the five criteria established above, and supported by proof-of-concept

evidence of responsiveness to geroprotective interventions. The data generated through the XPRIZE Healthspan competition itself represent a valuable opportunity to develop and validate such proxies, provided that biospecimen and immunoprofiling are embedded in trial design from the outset. Defining the minimally useful magnitude and time-scale of responsiveness—an important consideration given that many biomarkers are identified by comparisons over years—should be an explicit priority for these efforts.

Challenges and limitations

Several important challenges and limitations must be acknowledged. First, longitudinal reference data are needed to establish adequate response thresholds to determine if meaningful restoration of immune function has been achieved with intervention. Without such benchmarks, interpreting change scores in the context of a one-year trial remains difficult. Interim approaches—such as age-referenced thresholds provided here—provide a pragmatic starting point, but generalizable criteria applicable across populations, trial contexts and specific measures remain to be established.

Second, the evaluation presented here identifies promising candidate biomarkers and assesses their current evidentiary status, but it does not qualify any measures as a validated surrogate endpoint. Such qualification would require an accepted clinical outcome assessment for aging trials and evidence across multiple successful interventions demonstrating a consistent dose–response relationship between biomarker change and clinical benefit—a bar that no current immune aging biomarker has yet met. Recent efforts, including those from the Biomarkers of Aging Consortium^{54–56}, have begun outlining practical considerations for advancing biomarkers along this continuum. Formal pathways exist for this purpose, such as the US Food and Drug Administration (FDA) [Biomarker Qualification Program](#), which requires submitting a structured evidentiary package demonstrating a biomarker reliably predicts a clinical outcome of interest within a specified context of use, and the European Medicines Agency's parallel qualification procedure. These are complementary to, rather than a substitute for, the upstream scientific work of establishing expert consensus, replication across independent cohorts and consistent linkage between biomarker change and clinical benefit across multiple interventions.

Third, the biomarkers in this framework are derived predominantly from peripheral blood or otherwise noninvasively or minimally invasively collected samples, reflecting the practical realities of clinical trial settings where scalable sampling is essential. While peripheral blood provides a tractable and reproducible window into some aspects of immune aging, it may insufficiently capture tissue-resident immune dynamics and organ-specific aging processes that are increasingly recognized as central to immune aging. Where feasible, complementary assessments, including those based on imaging or emerging technologies, may provide supportive evidence.

Fourth, heterogeneity in technical capacity and resources across global clinical trial centers presents a practical challenge for implementation. A formally tiered biomarker framework distinguishing core, extended and exploratory panels is conceptually attractive, and one the field should work toward as evidence matures. However, the current evidentiary base does not yet support principled stratification, as the relative priority of individual biomarkers remains context dependent and feasibility will vary considerably across settings. As a step toward greater operational clarity, we provide a SOP workflow to guide implementation across sites (Fig. 2d), which we anticipate will serve as a practical scaffold for developing more formally tiered recommendations as longitudinal and multisite data accumulate.

Finally, the landscape of immune age biomarkers is evolving rapidly, with new measures and clinical validation data emerging in peer-reviewed literature on an almost weekly basis. It is likely that some measures were inadvertently overlooked or published during the development of this report. This pace of development is ultimately a

Table 2 | Roadmap for clinical immune aging biomarker development

Barrier	Action	Who should act	Existing levers	Needed developments
Functional assays are highly informative, but difficult to scale in multicenter trials	Develop validated proxies or simplified readouts of functional capacity	Researchers, funders, industry	Ex vivo stimulation assays; rich immunoprofiling cohorts	Proxy biomarkers validated against function; scalable trial-compatible workflows
Many assays and workflows remain labor intensive and costly	Invest in automation for processing and immunoprofiling or quantification; proxy biomarker development	Industry, technology developers, funders	Cytometry or -omics platforms in research labs and core facilities	Automated, high-throughput pipelines; robotics for sample handling; standardized QC protocols
Field's reliance on individual biomarker molecules risks poor generalizability	Build cost-effective panels or composite signatures, ideally capturing innate, adaptive and functional dimensions	Researchers, diagnostic companies	Multiplex panels and clocks	Cost-effective multidimensional panels; benchmarking across populations
Most biomarkers are evaluated cross-sectionally, limiting ability to capture trajectories and resilience	Establish longitudinal, multi-omics cohorts with repeat sampling	Funders, consortia, biobanks	Some existing cohorts with repeat immune measures	Large-scale longitudinal immune aging cohorts; harmonized follow-up and clinical endpoints
Unclear whether biomarker changes reflect adaptation or damage	Use mechanistic studies and intervention trials to clarify causal pathways	Researchers, clinical trialists	Vaccination and infection challenge studies	Integration with outcomes; intervention studies directly linking biomarkers to immune resilience
Protocols, assays and analytic pipelines differ widely across sites	Harmonize SOPs, develop QC frameworks, and engage regulators early for trial readiness	Researchers, regulators, consortia	Laboratory SOPs, early consortia efforts	Cross-site benchmarking; regulatory partnerships for immune biomarkers
Validation often restricted to narrow demographic or disease groups	Ensure global representation across age, sex, ancestry and health status	Funders, consortia, policymakers	Large biobanks	Coordinated international consortia; harmonization of datasets

sign of the field's momentum, and underscores the need for continued surveillance and iterative updating of candidate biomarker lists as trials progress and new evidence accumulates. The data generated through initiatives such as XPRIZE Healthspan, if leveraged systematically, could meaningfully accelerate this process, transforming early-stage competition trials into a platform for the next generation of validated immune aging endpoints.

A path toward actionable immune aging biomarkers

Despite rapid progress, few immune aging biomarkers currently meet the stringent criteria needed for clinical translation. Some of the most advanced biomarker candidates and functional ex vivo assays show promise in capturing multidimensional aspects of immune function and resilience. To realize their potential, coordinated efforts are required to address outstanding challenges and accelerate their adoption in trials and practice. Below, we outline a set of priorities that we see as urgent, defining a clear path for progress and alignment in the field (Table 2).

Functional proxies

Functional assays (for example, ex vivo stimulation, vaccination challenges) provide some of the most direct and biologically meaningful measures of immune resilience. However, their complexity and lack of standardized assays make them impractical for multicenter trials. Developing validated proxies or simplified readouts that retain functional information, but are compatible with large-scale settings, will be critical for clinical application. Candidate proxies could be benchmarked against existing functional assays in deeply phenotyped cohorts, creating scalable workflows that bridge mechanistic richness and trial feasibility.

Scalability and automation

Many immunoprofiling workflows remain costly, labor intensive and operator dependent. Advances in automation, from robotic handling to standardized high-throughput cytometry and omics pipelines, offer a way forward. Moving from bespoke research protocols to robust, quality-controlled pipelines will require coordinated investment across funders, technology developers and industry, together with harmonized quality-control standards to ensure site-to-site comparability.

Embracing multidimensionality

The field's past reliance on individual molecules such as IL-6 or CRP, risks poor generalizability. Integrative approaches combining innate, adaptive and functional dimensions of immune aging provide greater interpretability and predictive power^{86,87}. Indeed, complementary aspects of immunity can be partly decoupled; for example, repertoire diversity and inflammatory status are hypothesized as 'two sides of the same coin' but the trajectories of changes in both systems track together (reviewed in ref. 16). This highlights the need for multidimensional composites that capture these axes in parallel. Multiplex panels and multidimensional immune 'clocks' are emerging, and the next step is to benchmark these across populations, identify cost-effective designs and define which combinations best stratify immune resilience and disease risk. Such approaches may also reveal distinct immunophenotypes of aging, enabling more tailored intervention strategies. Large-scale studies, such as the recent *Nature* atlas of immune aging⁶⁰, demonstrate the power of multidimensional, comprehensive, population-level profiling to reveal conserved cellular and molecular signatures of aging. Such work provides essential reference trajectories and supports the development of clinically meaningful immune aging biomarkers capable of guiding personalized intervention strategies.

Longitudinal and multimodal cohorts

Most biomarkers so far have been assessed cross-sectionally, limiting their ability to capture trajectories or resilience. Large-scale longitudinal cohorts with repeated sampling and harmonized endpoints are urgently needed to define temporal dynamics and strengthen links to clinical outcomes. While longitudinal design is one of the main priorities, integrating multiple modalities within the same populations, such as immune cell composition, proteomics and DNA methylation, is an additional forward-looking priority that further enables comparison across layers and provides a more comprehensive view of immune aging.

Understanding mechanisms

A persistent conceptual barrier is distinguishing adaptive or allostatic responses from damage-related changes. While mechanistic understanding is not essential to a useful biomarker, many immune

alterations may represent compensatory mechanisms rather than pathological aging, and biomarkers that fail to capture this nuance risk misinterpretation. The IL-6 paradox, in which the same cytokine can both be adaptive and deleterious depending on context⁸⁸, illustrates the importance of this distinction. Embedding biomarker measurements in mechanistic studies and intervention trials is therefore important to clarify causal pathways.

Diversity and generalizability

Validation to date has often been limited to narrow demographic or disease groups. Expanding representation across age, sex, ancestry and health status will be vital to ensure generalizability and equity in biomarker application.

Several resources already exist to support these endeavors. These include cohorts with repeated immunoprofiling, as well as new initiatives expanding profiling, automation or scaling efforts. We summarize what we believe are the most relevant in Supplementary Table 1.

Conclusion

Immune aging is measurable and modifiable. With the above framework, we establish a systematic approach to selecting and validating immune aging biomarkers for geroscience clinical trials, addressing a critical gap in the field. As a call to action, we recommend clinicians to move beyond single inflammatory markers toward multidimensional composite measures that better capture immune fitness. Trialists should incorporate complementary biomarkers spanning inflammatory, senescence and functional dimensions with established validity and feasibility for serial assessment. Researchers should prioritize scalable omics-based proxies, standardization and longitudinal validation linking biomarker changes to clinically meaningful outcomes.

Realizing the full potential of immune aging biomarkers requires coordinated action across the translational ecosystem. Priority investments include developing scalable functional proxies, automating complex workflows, establishing longitudinal deeply phenotyped cohorts with diverse populations and harmonizing protocols across sites to enable regulatory acceptance. Although no single immune aging biomarker currently meets all criteria for a validated surrogate endpoint, the convergence of multidimensional immunoprofiling, longitudinal study designs and real-world trial initiatives positions the field to establish clinically meaningful immune aging endpoints within the next decade.

References

- Kuo, P.-L. et al. Longitudinal phenotypic aging metrics in the Baltimore longitudinal study of aging. *Nat. Aging* **2**, 635–643 (2022).
- Kennedy, B. K. et al. Geroscience: linking aging to chronic disease. *Cell* **159**, 709–713 (2014).
- Justice, J. N. XPRIZE healthspan is a global competition to restore function. *Nat. Aging* **4**, 165–166 (2024).
- López-Otín, C., Blasco, M. A., Partridge, L., Serrano, M. & Kroemer, G. Hallmarks of aging: an expanding universe. *Cell* **186**, 243–278 (2023).
- Yousefzadeh, M. J. et al. An aged immune system drives senescence and ageing of solid organs. *Nature* **594**, 100–105 (2021).
- Bartleson, J. M. et al. SARS-CoV-2, COVID-19 and the aging immune system. *Nat. Aging* **1**, 769–782 (2021).
- Lord, J. M. et al. Accelerated immune ageing is associated with COVID-19 disease severity. *Immun. Ageing* **21**, 6 (2024).
- Wu, F. et al. Immunological biomarkers of aging. *J. Immunol.* **214**, 889–902 (2025).
- Sender, R. et al. The total mass, number, and distribution of immune cells in the human body. *Proc. Natl Acad. Sci. USA* **120**, e2308511120 (2023).
- Terekhova, M., Bohacova, P. & Artyomov, M. N. Human immune aging. *Immunity* **58**, 2646–2669 (2025).
- Doherty, T. M., Weinberger, B., Didierlaurent, A. & Lambert, P.-H. Age-related changes in the immune system and challenges for the development of age-specific vaccines. *Ann. Med.* **57**, 2477300 (2025).
- Kulkarni, U. et al. Excessive neutrophil levels in the lung underlie the age-associated increase in influenza mortality. *Mucosal Immunol.* **12**, 545–554 (2019).
- Chen, J., Kelley, W. J. & Goldstein, D. R. Role of aging and the immune response to respiratory viral infections: potential implications for COVID-19. *J. Immunol.* **205**, ji2000380 (2020).
- Smith, C. A., Kulkarni, U., Chen, J. & Goldstein, D. R. Influenza virus inoculum volume is critical to elucidate age-dependent mortality in mice. *Aging Cell* **18**, e12893 (2019).
- Franceschi, C., Garagnani, P., Parini, P., Giuliani, C. & Santoro, A. Inflammaging: a new immune–metabolic viewpoint for age-related diseases. *Nat. Rev. Endocrinol.* **14**, 576–590 (2018).
- Fulop, T. et al. Immunosenescence and inflamm-aging as two sides of the same coin: friends or foes? *Front. Immunol.* **8**, 1960 (2018).
- Ferrucci, L. & Fabbri, E. Inflammaging: chronic inflammation in ageing, cardiovascular disease, and frailty. *Nat. Rev. Cardiol.* **15**, 505–522 (2018).
- Mogilenko, D. A., Shchukina, I. & Artyomov, M. N. Immune ageing at single-cell resolution. *Nat. Rev. Immunol.* **22**, 484–498 (2022).
- Weyand, C. M. & Goronzy, J. J. Aging of the immune system. mechanisms and therapeutic targets. *Ann. Am. Thorac. Soc.* **13**, S422–S428 (2016).
- Lynch, H. E. et al. Thymic involution and immune reconstitution. *Trends Immunol.* **30**, 366–373 (2009).
- Elyahu, Y. & Monsonego, A. Thymus involution sets the clock of the aging T-cell landscape: Implications for declined immunity and tissue repair. *Ageing Res. Rev.* **65**, 101231 (2021).
- Spadaro, O. et al. Caloric restriction in humans reveals immunometabolic regulators of health span. *Science* **375**, 671–677 (2022).
- Yang, H., Youm, Y. -H. & Dixit, V. D. Inhibition of thymic adipogenesis by caloric restriction is coupled with reduction in age-related thymic involution. *J. Immunol.* **183**, 3040–3052 (2009).
- Su, T.-Y. et al. Aging is associated with functional and molecular changes in distinct hematopoietic stem cell subsets. *Nat. Commun.* **15**, 7966 (2024).
- Beerman, I. et al. Functionally distinct hematopoietic stem cells modulate hematopoietic lineage potential during aging by a mechanism of clonal expansion. *Proc. Natl Acad. Sci. USA* **107**, 5465–5470 (2010).
- Jaiswal, S. & Ebert, B. L. Clonal hematopoiesis in human aging and disease. *Science* **366**, eaan4673 (2019).
- Bouaziz, H. et al. Clonal hematopoiesis is associated with protection from Alzheimer’s disease. *Nat. Med.* **29**, 1662–1670 (2023).
- Masters, A. R. et al. Assessment of lymph node stromal cells as an underlying factor in age-related immune impairment. *J. Gerontol. A Biol. Sci. Med. Sci.* **74**, 1734–1743 (2019).
- Gao, X., Zhang, J. & Tamplin, O. J. The aging hematopoietic stem cell niche: a mini review. *Front. Hematol.* **4**, 1525132 (2025).
- Ross, J. B. et al. Depleting myeloid-biased haematopoietic stem cells rejuvenates aged immunity. *Nature* **628**, 162–170 (2024).
- Mogilenko, D. A. et al. Comprehensive profiling of an aging immune system reveals clonal GZMK⁺ CD8⁺ T cells as conserved hallmark of inflammaging. *Immunity* **54**, 99–115 (2021).

32. Mol, J. de, Kuiper, J., Tsiantoulas, D. & Foks, A. C. The dynamics of B cell aging in health and disease. *Front. Immunol.* **12**, 733566 (2021).
33. Mittelbrunn, M. & Kroemer, G. Hallmarks of T cell aging. *Nat. Immunol.* **22**, 687–698 (2021).
34. Sun, X. et al. Longitudinal analysis reveals age-related changes in the T cell receptor repertoire of human T cell subsets. *J. Clin. Invest.* **132**, e158122 (2022).
35. Park, M. D. et al. Restoring resident tissue macrophages to combat aging and cancer. *Nat. Aging* **5**, 1383–1392 (2025).
36. Adé, K. et al. Inflammation drives age-induced loss of tissue resident macrophages. Preprint at *bioRxiv* <https://doi.org/10.1101/2022.10.02.510513> (2022).
37. Krishnarajah, S. et al. Single-cell profiling of immune system alterations in lymphoid, barrier and solid tissues in aged mice. *Nat. Aging* **2**, 74–89 (2022).
38. Mesa, K. R. et al. Niche-specific dermal macrophage loss promotes skin capillary ageing. *Nature* **648**, 173–181 (2025).
39. Keenan, C. R. & Allan, R. S. Epigenomic drivers of immune dysfunction in aging. *Aging Cell* **18**, e12878 (2019).
40. Wang, K. et al. Epigenetic regulation of aging: implications for interventions of aging and diseases. *Signal Transduct. Target. Ther.* **7**, 374 (2022).
41. Beerman, I. et al. Proliferation-dependent alterations of the DNA methylation landscape underlie hematopoietic stem cell aging. *Cell Stem Cell* **12**, 413–425 (2013).
42. Yanai, H. et al. DNA methylation drives hematopoietic stem cell aging phenotypes after proliferative stress. *GeroScience* <https://doi.org/10.1007/s11357-024-01360-4> (2024).
43. Bruhn-Olszewska, B. et al. The effects of loss of Y chromosome on male health. *Nat. Rev. Genet.* **26**, 320–335 (2025).
44. Roberts, A. L. et al. Age acquired skewed X chromosome inactivation is associated with adverse health outcomes in humans. *eLife* **11**, e78263 (2022).
45. Fali, T. et al. New insights into lymphocyte differentiation and aging from telomere length and telomerase activity measurements. *J. Immunol.* **202**, 1962–1969 (2019).
46. Karimian, K. et al. Human telomere length is chromosome end-specific and conserved across individuals. *Science* **384**, 533–539 (2024).
47. Smoom, R., Lichtental, D., Kaestner, K. H. & Tzfati, Y. The house mouse maintains constant telomere length throughout life. *Nucleic Acids Res.* **53**, gkaf830 (2025).
48. McGuire, P. J. Mitochondrial dysfunction and the aging immune system. *Biology* **8**, 26 (2019).
49. Bektas, A. et al. Age-associated changes in human CD4⁺ T cells point to mitochondrial dysfunction consequent to impaired autophagy. *Aging* **11**, 9234–9263 (2019).
50. Thomas, R., Wang, W. & Su, D. -M. Contributions of age-related thymic involution to immunosenescence and inflammaging. *Immun. Ageing* **17**, 2 (2020).
51. Verbrugge, L. M. & Jette, A. M. The disablement process. *Soc. Sci. Med.* **38**, 1–14 (1994).
52. Reid, K. F. & Fielding, R. A. Skeletal muscle power. *Exerc. Sport Sci. Rev.* **40**, 4–12 (2012).
53. Kritchevsky, S. B. & Justice, J. N. Testing the geroscience hypothesis: early days. *J. Gerontol. A Biol. Sci. Med. Sci.* **75**, 99–101 (2020).
54. Moqri, M. et al. Biomarkers of aging for the identification and evaluation of longevity interventions. *Cell* **186**, 3758–3775 (2023).
55. Moqri, M. et al. Validation of biomarkers of aging. *Nat. Med.* **30**, 360–372 (2024).
56. Herzog, C. M. S. et al. Challenges and recommendations for the translation of biomarkers of aging. *Nat. Aging* **4**, 1372–1383 (2024).
57. Ahuja, S. K. et al. Immune resilience despite inflammatory stress promotes longevity and favorable health outcomes including resistance to infection. *Nat. Commun.* **14**, 3286 (2023).
58. Manoharan, M. S. et al. The 15-year survival advantage: immune resilience as a salutogenic force in healthy aging. *Aging Cell* **24**, e70063 (2025).
59. Franck, M. et al. Nonuniversality of inflammaging across human populations. *Nat. Aging* **5**, 1471–1480 (2025).
60. Gong, Q. et al. Multi-omic profiling reveals age-related immune dynamics in healthy adults. *Nature* **648**, 696–706 (2025).
61. Weng, N., Akbar, A. N. & Goronzy, J. CD28⁺ T cells: their role in the age-associated decline of immune function. *Trends Immunol.* **30**, 306–312 (2009).
62. Liu, Y. et al. Expression of p16INK4a in peripheral blood T-cells is a biomarker of human aging. *Aging Cell* **8**, 439–448 (2009).
63. Hillary, R. F. et al. Blood-based epigenome-wide analyses of chronic low-grade inflammation across diverse population cohorts. *Cell Genom.* **4**, 100544 (2024).
64. Dawoud, A., Green, L. & Rackham, O. Loss of chromosome Y shapes the immune cell fate with aging in men. *Aging Cell* **25**, e70528 (2026).
65. Alpert, A. et al. A clinically meaningful metric of immune age derived from high-dimensional longitudinal monitoring. *Nat. Med.* **25**, 487–495 (2019).
66. Sayed, N. et al. An inflammatory aging clock (iAge) based on deep learning tracks multimorbidity, immunosenescence, frailty and cardiovascular aging. *Nat. Aging* **1**, 598–615 (2021).
67. Goyani, P., Christodoulou, R. & Vassiliou, E. Immunosenescence: aging and immune system decline. *Vaccines* **12**, 1314 (2024).
68. Rodrigues, L. P. et al. Hallmarks of aging and immunosenescence: connecting the dots. *Cytokine Growth Factor Rev.* **59**, 9–21 (2021).
69. Raza, K. et al. Specific features of immune ageing are detected in the earliest stages in rheumatoid arthritis development. *eBioMedicine* **119**, 105900 (2025).
70. Foster, M. A. et al. Investigating the potential of a prematurely aged immune phenotype in severely injured patients as predictor of risk of sepsis. *Immun. Ageing* **19**, 60 (2022).
71. Bröde, P. et al. Calibrating a comprehensive immune age metric to analyze the cross sectional age-related decline in cardiorespiratory fitness. *Biology* **11**, 1576 (2022).
72. Gajewski, P. D. et al. A systematic analysis of biological, sociodemographic, psychosocial, and lifestyle factors contributing to work ability across the working life span: cross-sectional study. *JMIR Form. Res.* **7**, e40818 (2023).
73. Sánchez-Sánchez, J. L. et al. Biological ageing acceleration and functional capacities across the lifespan in the INSPIRE-T cohort. *J. Cachexia Sarcopenia Muscle* **16**, e70046 (2025).
74. Wang, R. Z. et al. Inflammatory age and its impact on age-related health in older Chinese adults. *Arch. Gerontol. Geriatr.* **125**, 105476 (2024).
75. Schmunk, L. J. et al. A novel framework to build saliva-based DNA methylation biomarkers: quantifying systemic chronic inflammation as a case study. *Aging Cell* **24**, e14444 (2025).
76. Schafer, M. J. et al. The senescence-associated secretome as an indicator of age and medical risk. *JCI Insight* **5**, e133668 (2020).
77. Cummings, S. R. et al. Biomarkers of cellular senescence and major health outcomes in older adults. *GeroScience* **47**, 3407–3415 (2025).
78. St Sauver, J. L. et al. Biomarkers of cellular senescence and risk of death in humans. *Aging Cell* <https://doi.org/10.1111/acer.14006> (2023).
79. Fielding, R. A. et al. Associations between biomarkers of cellular senescence and physical function in humans: observations from the lifestyle interventions for elders (LIFE) study. *GeroScience* <https://doi.org/10.1007/s11357-022-00685-2> (2022).

80. Mielke, M. M. et al. Biomarkers of cellular senescence predict risk of mild cognitive impairment: Results from the lifestyle interventions for elders (LIFE) study. *J. Nutr. Health Aging* **29**, 100529 (2025).
81. Centers for Disease Control. QuickStats: death rates from influenza and pneumonia among persons aged ≥65 years, by sex and age group — National Vital Statistics System, United States, 2018. *MMWR Morb. Mortal. Wkly Rep.* **69**, 1470 (2020).
82. Montgomery, R. R. Age-related alterations in immune responses to West Nile virus infection. *Clin. Exp. Immunol.* **187**, 26–34 (2016).
83. Abel, L. & Casanova, J. -L. Human determinants of age-dependent patterns of death from infection. *Immunity* **57**, 1457–1465 (2024).
84. Mannick, J. B. et al. Targeting the biology of ageing with mTOR inhibitors to improve immune function in older adults: phase 2b and phase 3 randomised trials. *Lancet Healthy Longev.* **2**, e250–e262 (2021).
85. Mannick, J. B. et al. TORC1 inhibition enhances immune function and reduces infections in the elderly. *Sci. Transl. Med.* **10**, eaaq1564 (2018).
86. Morrisette-Thomas, V. et al. Inflamm-aging does not simply reflect increases in pro-inflammatory markers. *Mech. Ageing Dev.* **139**, 49–57 (2014).
87. Furman, D. et al. Chronic inflammation in the etiology of disease across the life span. *Nat. Med.* **25**, 1822–1832 (2019).
88. Franceschi, C. et al. Inflamm-aging: an evolutionary perspective on immunosenescence. *Ann. N.Y. Acad. Sci.* **908**, 244–254 (2000).
89. Diniz, B. S. et al. Association of molecular senescence markers in late-life depression with clinical characteristics and treatment outcome. *JAMA Netw. Open* **5**, e2219678 (2022).
90. Dioum, E. H. M. et al. Oats lower age-related systemic chronic inflammation (iAge) in adults at risk for cardiovascular disease. *Nutrients* **14**, 4471 (2022).
91. Fuentealba, M. et al. Multi-omics analysis reveals biomarkers that contribute to biological age rejuvenation in response to single-blinded randomized placebo-controlled therapeutic plasma exchange. *Aging Cell* **24**, e70103 (2025).
92. Nimmerjahn, F., Vidarsson, G. & Cragg, M. S. Effect of posttranslational modifications and subclass on IgG activity: from immunity to immunotherapy. *Nat. Immunol.* **24**, 1244–1255 (2023).
93. Vinicki, M. et al. Effects of testosterone and metformin on the GlycanAge index of biological age and the composition of the IgG glycome. *GeroScience* **47**, 1777–1788 (2025).
94. Rapčan, B. et al. Glycan clock of ageing—analytical precision and time-dependent inter- and i-individual variability. *GeroScience* **46**, 5781–5796 (2024).
95. Sun, S. et al. Immunoglobulin G N-glycosylation predicts long-term risk of ischemic stroke: a nested case-control study. *J. Transl. Med.* **23**, 1175 (2025).
96. Wang, H. et al. Immunoglobulin G N-glycome as a biomarker of mortality risk in *Escherichia coli* induced sepsis. *Front. Immunol.* **16**, 1532145 (2025).
97. Hoshi, R. A. et al. N-Glycosylation profiles of immunoglobulin G and future cardiovascular events. *Circ. Res.* **134**, e3–e14 (2024).
98. Burel, J. G. et al. IgG glycosylation associates with risk of progression from latent to active tuberculosis. *J. Infect.* **88**, 106115 (2024).
99. Song, M., Graubard, B. I., Rabkin, C. S. & Engels, E. A. Neutrophil-to-lymphocyte ratio and mortality in the United States general population. *Sci. Rep.* **11**, 464 (2021).
100. Dillon, K., Goodman, Z. T., Kaur, S. S., Levin, B. & McIntosh, R. Neutrophil-to-Lymphocyte ratio amplifies the effects of aging on decrements in grip strength and its functional neural underpinnings. *J. Gerontol. A Biol. Sci. Med. Sci.* **78**, 882–889 (2023).
101. Yoshida, Y., Iwasa, H., Kim, H. & Suzuki, T. Association between neutrophil-to-lymphocyte ratio and physical function in older adults: a community-based cross-sectional study in Japan. *Int. J. Environ. Res. Public Health* **19**, 8996 (2022).
102. Willik et al. Balance between innate versus adaptive immune system and the risk of dementia: a population-based cohort study. *J. Neuroinflammation* **16**, 68 (2019).
103. Bhat, T. et al. Neutrophil to lymphocyte ratio and cardiovascular diseases: a review. *Expert Rev. Cardiovasc. Ther.* **11**, 55–59 (2013).
104. Ethier, J. -L., Desautels, D., Templeton, A., Shah, P. S. & Amir, E. Prognostic role of neutrophil-to-lymphocyte ratio in breast cancer: a systematic review and meta-analysis. *Breast Cancer Res.* **19**, 2 (2017).
105. Jacobs, T. et al. The neutrophil to lymphocyte ratio associates with markers of Alzheimer’s disease pathology in cognitively unimpaired elderly people. *Immun. Ageing* **21**, 32 (2024).
106. Neupane, R. et al. Immune disorder in atherosclerotic cardiovascular disease—clinical implications of using circulating T-cell subsets as biomarkers. *Circ. J.* **83**, 1431–1438 (2019).
107. Yang, Q. -K. et al. CONUT score or/and peripheral blood CD4⁺/CD8⁺ ratio-based web dynamic nomograms to predict the individualized survival of patients with advanced osteosarcoma. *Cancer Manag. Res.* **12**, 4193–4208 (2020).
108. Adriaensen, W. et al. CD4:8 ratio >5 is associated with a dominant naive T-cell phenotype and impaired physical functioning in CMV-seropositive very elderly people: results from the BELFRAIL study. *J. Gerontol. A Biol. Sci. Med. Sci.* **70**, 143–154 (2015).
109. Formiga, F., Ferrer, A., Padros, G., Contra, A. & Pujol, R. Inverted CD4:CD8 ratio is not associated with three-year mortality in a sample of community-dwelling oldest old: The Octabaix immune study. *J. Nutr. Health Aging* **18**, 425–428 (2014).
110. Ragab, A. A. Y. et al. Immune cell phenotypes and mortality in the framingham heart study. *Immun. Ageing* **21**, 37 (2024).
111. Fang, Y. et al. Circulating immune cell phenotypes are associated with age, sex, CMV, and smoking status in the framingham heart study offspring participants. *Aging* **15**, 3939–3966 (2023).
112. Ramasubramanian, R. et al. Evaluation of T-cell aging-related immune phenotypes in the context of biological aging and multimorbidity in the Health and Retirement Study. *Immun. Ageing* **19**, 33 (2022).
113. Wielscher, M. et al. DNA methylation signature of chronic low-grade inflammation and its role in cardio-respiratory diseases. *Nat. Commun.* **13**, 2408 (2022).
114. McGee, K. C. et al. A combination nutritional supplement reduces DNA methylation age only in older adults with a raised epigenetic age. *GeroScience* **46**, 4333–4347 (2024).
115. Reading, C. L. et al. An exploratory analysis of bezisterim treatment associated with decreased biological age acceleration, and improved clinical measure and biomarker changes in mild-to-moderate probable Alzheimer’s disease. *Front. Neurosci.* **19**, 1516746 (2025).
116. Middelkamp, V. & Kekäläinen, E. Measuring thymic output across the human lifespan: a critical challenge in laboratory medicine. *GeroScience* **47**, 6797–6806 (2025).
117. Lakhonina, N. et al. T cell receptor rearrangement excision circles (TREC) study as an approach to ‘in vivo’ thymus gland function investigation. *Arthritis Res. Ther.* **14**, P37 (2012).
118. Mikhael, N. L. & Elsorady, M. Clinical significance of T cell receptor excision circle (TREC) quantitation after allogeneic HSCT. *Blood Res.* **54**, 274–281 (2019).
119. Sandhar, B. et al. Heterogeneity of thymic output in the elderly and its association with sex and smoking. *JCI Insight* **10**, e189008 (2025).

120. Sparks, R. et al. A unified metric of human immune health. *Nat. Med.* **30**, 2461–2472 (2024).
121. Chen, X. et al. Impact of infection on healthcare costs and clinical outcomes in elderly hospitalized patients with multimorbidity. *Heliyon* **10**, e31560 (2024).
122. Lord, J. M. The effect of aging of the immune system on vaccination responses. *Hum. Vacc. Immunother.* **9**, 1364–1367 (2013).
123. Collier, D. A. et al. Age-related immune response heterogeneity to SARS-CoV-2 vaccine BNT162b2. *Nature* **596**, 417–422 (2021).
124. Goodwin, K., Viboud, C. & Simonsen, L. Antibody response to influenza vaccination in the elderly: a quantitative review. *Vaccine* **24**, 1159–1169 (2006).
125. Semelka, C. T. et al. Frailty and COVID-19 mRNA vaccine antibody response in the COVID-19 community research partnership. *J. Gerontol. A Biol. Sci. Med. Sci.* **77**, 1366–1370 (2022).
126. Macdonald, C. et al. Association between antibody responses post-vaccination and severe COVID-19 outcomes in Scotland. *NPJ Vaccines* **9**, 107 (2024).
127. Martin, D. E. et al. The effect of metformin on influenza vaccine responses in nondiabetic older adults: a pilot trial. *Immun. Ageing* **20**, 18 (2023).
128. Agrawal, A. et al. Altered innate immune functioning of dendritic cells in elderly humans: a role of phosphoinositide 3-kinase-signaling pathway. *J. Immunol.* **178**, 6912–6922 (2007).
129. Merani, S., Kuchel, G. A., Kleppinger, A. & McElhaney, J. E. Influenza vaccine-mediated protection in older adults: impact of influenza infection, cytomegalovirus serostatus and vaccine dosage. *Exp. Gerontol.* **107**, 116–125 (2018).
130. McElhaney, J. E. et al. T cell responses are better correlates of vaccine protection in the elderly. *J. Immunol.* **176**, 6333–6339 (2006).
131. Wells, S. B. et al. Multimodal profiling reveals tissue-directed signatures of human immune cells altered with age. *Nat. Immunol.* **26**, 1612–1625 (2025).
132. Lam, N. et al. Asynchronous aging and turnover of human circulating and tissue-resident memory T cells across sites. *Immunity* **58**, 2271–2288 (2025).
133. Pawelec, G. Age and immunity: what is “immunosenescence”? *Exp. Gerontol.* **105**, 4–9 (2018).
134. Lee, K. -A., Flores, R. R., Jang, I. H., Saathoff, A. & Robbins, P. D. Immune senescence, immunosenescence and aging. *Front. Aging* **3**, 900028 (2022).
135. Wrona, M. V., Ghosh, R., Coll, K., Chun, C. & Yousefzadeh, M. J. The 3 I's of immunity and aging: immunosenescence, inflammaging, and immune resilience. *Front. Aging* **5**, 1490302 (2024).
136. Goronzy, J. J. & Weyand, C. M. Understanding immunosenescence to improve responses to vaccines. *Nat. Immunol.* **14**, 428–436 (2013).
137. Moskalev, A., Stambler, I. & Caruso, C. Innate and adaptive immunity in aging and longevity: the foundation of resilience. *Aging Dis.* **11**, 1363–1373 (2020).
138. Verster, J. C., Kraneveld, A. D. & Garssen, J. The assessment of immune fitness. *J. Clin. Med.* **12**, 22 (2022).
139. Dolan, M., Libby, K. A., Ringel, A. E., Galen, P. van & McAllister, S. S. Ageing, immune fitness and cancer. *Nat. Rev. Cancer* **25**, 848–872 (2025).
140. Dantzer, R., Cohen, S., Russo, S. J. & Dinan, T. G. Resilience and immunity. *Brain Behav. Immun.* **74**, 28–42 (2018).
141. Montano, M. Immune resilience: rewriting the rules of healthy aging. *Aging Cell* **24**, e70089 (2025).
142. Baessler, A. & Vignali, D. A. A. T cell exhaustion. *Ann. Rev. Immunol.* **42**, 179–206 (2024).
143. Slaets, H., Veeningen, N., Keizer, P. L. J. de, Hellings, N. & Hendrix, S. Are immunosenescent T cells really senescent? *Aging Cell* **23**, e14300 (2024).
144. Suryadevara, V. et al. SenNet recommendations for detecting senescent cells in different tissues. *Nat. Rev. Mol. Cell Biol.* **25**, 1001–1023 (2024).
145. Beltra, J. -C. et al. Developmental relationships of four exhausted CD8⁺ T cell subsets reveals underlying transcriptional and epigenetic landscape control mechanisms. *Immunity* **52**, 825–841 (2020).
146. Wherry, E. J. & Kurachi, M. Molecular and cellular insights into T cell exhaustion. *Nat. Rev. Immunol.* **15**, 486–499 (2015).
147. Márquez, E. J. et al. Sexual-dimorphism in human immune system aging. *Nat. Commun.* **11**, 751 (2020).
148. Franceschi, C. & Campisi, J. Chronic inflammation (inflammaging) and its potential contribution to age-associated diseases. *J. Gerontol. A Biol. Sci. Med. Sci.* **69**, S4–S9 (2014).
149. Ridker, P. M. et al. Antiinflammatory therapy with canakinumab for atherosclerotic disease. *New Engl. J. Med.* **377**, 1119–1131 (2017).

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Author contributions

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Competing interests

XPRIZE Healthspan is offered with support from co-title sponsors Hevolution Foundation and C. Wilson, founder and chairman of SOLVE FSHD, and Pharmaceutical Sponsor GSK. O.O.A. is a cofounder of Terrain Bio, Transit Therapeutics and Monet Therapeutics, and has filed intellectual property related to CRISPR-based genome editing and aging/rejuvenation. D.E.M.-H. is a cofounder and shareholder at Chronomics, a company focused on multi-omics and epigenetic biomarker development and diagnostic infrastructure. A.N.A. is a judge of the XPrize Healthspan scheme, chair of the scientific advisory board of the Walgreens Boots Alliance and University of Manchester collaborative skin initiative, and chair of the advisory board for the Singapore Ethnic Ageing grant at Singapore A*STAR Skin Research Labs. S.S.S.-O. holds equity and consults for CytoReason. W.W. is a cofounder of the company Cygenia, which provides services for epigenetic clocks to other scientists. D.F. is a founder of Edifice Health, a company that commercializes the iAge test. The other authors declare no competing interests.

Additional information

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